

Day 7 — Sāyana vs Nirayana

Module: Movement of the Sun · **Band:** Middle · **Time:** 45 min

Learning objective

By the end of this lesson, students will distinguish the **sāyana** (tropical) and **nirayana** (sidereal) zodiacs, explain that the gap between them — the **ayanāṁśa** — is $\sim 24^\circ$ today, and link this gap to **precession** (the slow wobble of Earth’s axis).

Materials

- Whiteboard
- Two transparent circles printed on overhead-projector film or paper-cutouts: each shows the 12 *rāśis* arranged around a circle
- A long printed strip showing the year (Jan to Dec) with the two solstice markers
- Student notebooks

Vocabulary introduced today

- *sāyana* (IAST: *sāyana*; lit. “with-ayana / tropical”) — the reference frame fixed to the equinoxes. The Sun enters *Meṣa* (Aries) on the **March equinox** by definition. The seasons stay locked to this system.
- *nirayana* (IAST: *nir-ayana*; lit. “without-ayana / sidereal”) — the reference frame fixed to the **fixed stars**. The Sun enters *Meṣa* when it physically passes a fixed point against the background stars.
- *ayanāṁśa* (IAST: *ayana-aṁśa*; lit. “precessional portion”) — the angular offset between the two systems, $\sim 24^\circ$ today.
- **Precession** — the slow wobble of Earth’s rotation axis. It traces a cone in space over $\sim 25,800$ years. Because of this wobble, the position of the equinoxes drifts among the stars at ~ 50 seconds of arc per year ($\sim 1^\circ$ per 72 years).

Lesson flow

Warm-up (5 min)

- Recall the puzzle: *Makara Saṁkrānti* is ~ 14 January but the winter solstice is ~ 21 December — a 24-day gap.
- Prompt: “24 days. Why 24?”
- Take 2 guesses. Park them. “Today we resolve it.”

Core (25 min)

1. **Two ways to define the start of the zodiac.**

Imagine the great circle of the ecliptic divided into 12 *rāśis*. **Where does *Meṣa* (the first *rāśi*) begin?**

- **Option A (sāyana / tropical):** *Meṣa* begins at **the March equinox point**. This makes the Sun’s position lock-step with the seasons. When the Sun is at 0° *Meṣa*, it’s the March equinox — always. Modern Western astronomy uses this.
- **Option B (nirayana / sidereal):** *Meṣa* begins at **a fixed point against the stars** (chosen near the star Spica / *Citrā* by Indian convention). When the Sun is at 0° *Meṣa*, it’s physically passing that star-anchored boundary. Indian traditional *jyotiṣa* uses this.

2. **Why do they differ?** Because of **precession**.

- Earth’s rotation axis isn’t perfectly fixed in space. It’s a slow wobble — like a spinning top tilting in a circle.
- One full wobble takes about **25,800 years**.
- The equinox point (where the Sun crosses the equator going north) is defined by the axis — so as the axis wobbles, the equinox point **drifts** against the background stars at about **50 seconds of arc per year**, or **1° per 72 years**.
- The drift goes *backwards* through the zodiac — from *Meṣa* into *Mīna*, then into *Kumbha*, and so on.

3. **The size of the gap right now.**

- The two systems were aligned approximately **1700 years ago** (in the early centuries CE).
- In that time, precession has drifted the equinox by **~24°** against the stars.
- So the *sāyana* zodiac and the *nirayana* zodiac are now **24° apart**.
- $24^\circ / 360^\circ \times 365.25 \text{ days} \approx 24.3 \text{ days}$. That’s the gap.
- **24-day gap = 24° offset = ~1700 years of precession**. That’s why *Makara Saṁkrānti* is 24 days *after* the winter solstice.

4. **Demo with the two transparencies.**

- Place the two *rāśi* circles on top of each other, both starting at the top.
- Slide one circle 24° clockwise. Now both circles still have *Meṣa*, *Vṛṣabha*, ... in order, but the boundaries are offset.
- “If the Sun is at 0° *Meṣa* in the bottom (*sāyana*) circle on March 21, where is it in the top (*nirayana*) circle?” — answer: it’s at 0° *Mīna* + 6° (since it’s offset 24° backward). It enters *Meṣa nirayana* ~24 days later, on ~April 14.

5. **A summary table:**

Event	<i>Sāyana</i> date	<i>Nirayana</i> date	Gap
Sun enters <i>Makara</i> / Capricorn	~21 December (= winter solstice)	~14 January (= Makara Saṁkrānti)	~24 days
Sun enters <i>Karka</i> / Cancer	~21 June (= summer solstice)	~16 July	~25 days
Sun enters	~21 March (= March equinox)	~14 April	~24

Event	<i>Sāyana</i> date	<i>Nirayana</i> date	Gap
<i>Meṣa</i> / Aries			days
Sun enters <i>Tulā</i> / Libra	~23 September (= September equinox)	~17 October	~24 days

6. **Which is “right”?** Neither — they’re answering different questions.

- *Sāyana* answers: “Where is the Sun relative to the seasons (equinoxes / solstices)?” — used by modern astronomy and the civil calendar.
- *Nirayana* answers: “Where is the Sun against the actual background stars?” — used by traditional Indian almanacs.
- **Both are internally consistent.** They diverge because the equinox drifts; whichever frame you fix, the other one moves.

Activity (10 min) — Compare-contrast worksheet

(See [activities/activity-03-sayana-nirayana.md](#) for the full sheet.)

In pairs, students fill a two-column table:

Question	<i>Sāyana</i> answer	<i>Nirayana</i> answer
Sun in <i>Meṣa</i> on ...	~21 March	~14 April
Aligned with seasons?	Yes (by definition)	Drifts over centuries
Aligned with stars?	Drifts over centuries	Yes (by definition)
Used by modern astronomy?	Yes	No
Used by Indian <i>pañcāṅga</i> ?	No	Yes
What changes over millennia?	Star background drifts	Season alignment drifts

Wrap-up (5 min)

- Exit ticket: “*Why is Makara Saṁkrānti on January 14 and not December 21? Answer in one sentence.*”
- Acceptable answer: “Because the *nirayana* (sidereal) zodiac, used for *Makara Saṁkrānti*, is offset from the *sāyana* (tropical) zodiac, used for the solstice, by ~24° due to precession over ~1700 years.”
- Preview Day 8: “*Tomorrow — build the project. Your rāśi-wheel will show BOTH systems.*”

Student-facing content

Two zodiacs, one sky.

System	Anchored to	Used by
<i>Sāyana</i> (tropical)	The equinoxes	Modern astronomy, Western horoscopes
<i>Nirayana</i> (sidereal)	The fixed stars	Indian traditional almanacs

The two were aligned ~1700 years ago. Earth’s slow axial wobble — **precession** — has drifted them apart by ~**24°** since. That gap is called the **ayanāmśa**.

$24^\circ \div 360^\circ \times 365.25 \text{ days} \approx \mathbf{24 \text{ days}}$. That’s why *Makara Saṁkrānti* (Jan 14, *nirayana*) is 24 days after the winter solstice (Dec 21, *sāyana*).

Homework

- Look up the *ayanāmśa* value for today on any astronomy website or *pañcāṅga* app. Most sources will quote either the *Lahiri* value (the Government of India standard) or the *Krishnamurti* value. They differ by a few minutes of arc. Note both.
- Write 4 sentences: explain to a younger sibling why their newspaper horoscope sign (Western, *sāyana*) might be one *rāśi* “off” from a traditional *pañcāṅga* (Indian, *nirayana*).

Differentiation

- One tier up:** When will the *sāyana* and *nirayana* zodiacs realign? Hint: when *ayanāmśa* completes another ~336° (= 25,800 yrs × 336/360). Approximately when? (Answer: in ~24,000 years.) What was the *ayanāmśa* in 1 CE? (Answer: very close to 0°.)
- One tier down:** Just remember: ~24 days difference. *Sāyana* = with the seasons. *Nirayana* = with the stars. Don’t worry about precession mechanics yet.

Teacher notes

- This is the densest day of the module. Plan to take questions — and to defer some to next class. The full theory of precession is the Day 8 *senior* topic; here we just name it.
- Some students will be upset that “the calendar is wrong.” Reframe: neither calendar is wrong. Each is right *within its frame*. The drift is a real astronomical phenomenon.
- If you have only 35 minutes, drop the worksheet and use the time for questions and the demo.

Citation(s) used in this lesson

- Precession rate (~50"/year, ~25,800-year period) is a modern astronomy standard value (see [sources.md](#)). Lahiri *ayanāmśa* is the Government of India standard adopted in the 1950s by the Calendar Reform Committee chaired by Meghnad Saha.